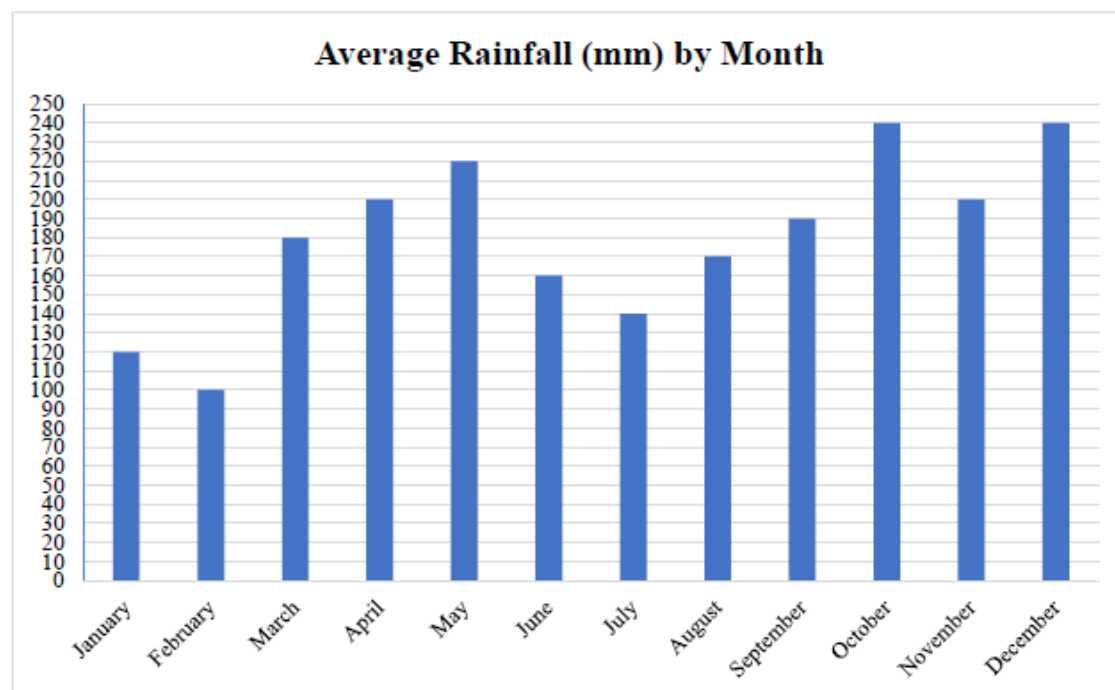


Singapore is looking at eliminating its reliance on imported water in the long run and you are tasked to determine the most cost-effective and environmentally responsible water supply plan using rainfall, NEWater and desalinated water to meet Singapore's demand for water.

Some information on the amount of rainfall and our local reservoirs is shown below:



- The amount of rainfall is measured by its depth in mm.
- Catchment area for local reservoirs is 180 km^2 .
- Runoff efficiency is 75% (i.e., only 75% of rainfall becomes reservoir water).
- Total reservoir storage capacity is 90 million m^3 .

The monthly water demand in Singapore is 110 million m^3 .

(a) Calculate the average monthly amount of rainfall (in mm).

[2]

(b) Show that the average monthly volume of water (in m^3) that can be collected from rainfall into the reservoirs is 24.3 million m^3 .

(Hint: $\text{Volume} = \text{Catchment area} \times \text{Height of rainfall}$)

[2]

- (c) Other than rainfall collected in reservoirs, Singapore also uses NEWater and desalinated water to meet its demand for water. Information on NEWater production and operation of desalination plant is shown below.
- Information on NEWater production and operation of desalination plant is shown below.

Source	Max Capacity (mil m ³ / month)	Cost per m ³	Energy Use (kWh/m ³)	CO ₂ Emissions (kg/m ³)
NEWater	30	\$1.30	0.5	0.2
Desalination	70	\$2.50	2.5	1.5

Electricity cost: \$0.26 per kWh (fixed)

There is a cap on total carbon emissions from NEWater production and desalination plant at 70 million kg CO₂ per month.

The table below shows the amount of penalty for carbon emissions above the cap.

Carbon Emissions (mil kg CO ₂)	Carbon Penalty
≤ 70	No penalty
70 < CO ₂ ≤ 90	\$0.50 per kg of CO ₂ emission
90 < CO ₂ ≤ 100	\$1.00 per kg of CO ₂ emission
> 100	\$2.00 per kg of CO ₂ emission

For desalinated volume above 50 million m³, there will be a penalty of \$0.50/m³ incurred on the volume exceeded.

The water supply options are shown in the table below:

Option	NEWater (m ³)	Desalination (m ³)
A	28 mil	57.7 mil
B	30 mil	55.7 mil
C	10 mil	75.7 mil
D	20 mil	45 mil

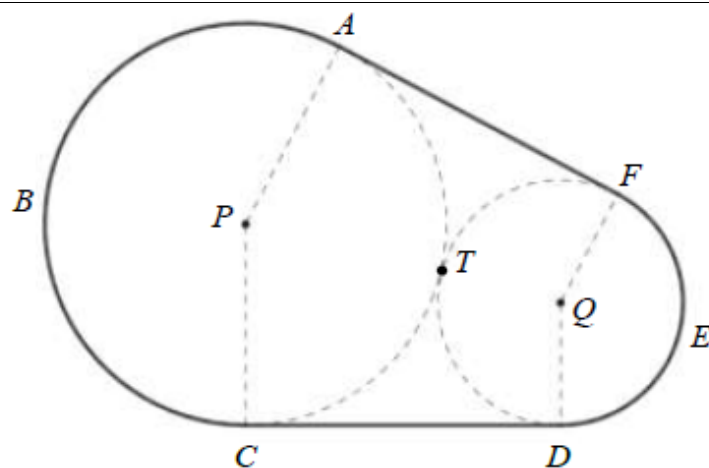
- (i) The monthly water demand in Singapore is 110 million m³.

By considering the monthly water demand in Singapore, the average monthly volume of rainfall and the maximum capacity of NEWater production and the desalination plant, which 2 options can be eliminated immediately? Explain your answer.

[1]

- (ii) Decide which option Singapore should choose and justify your answer clearly.

[6]



The diagram shows the floor plan of a room.

The room is formed by joining two circles, $ABCT$, centre P and $DEFT$, centre Q , at T .

AF and CD are tangents to both circles.

The perimeter of the room is bounded by the arc ABC , the arc DEF , the tangent AF and the tangent CD .

The radius of the larger circle is 7.2 m.

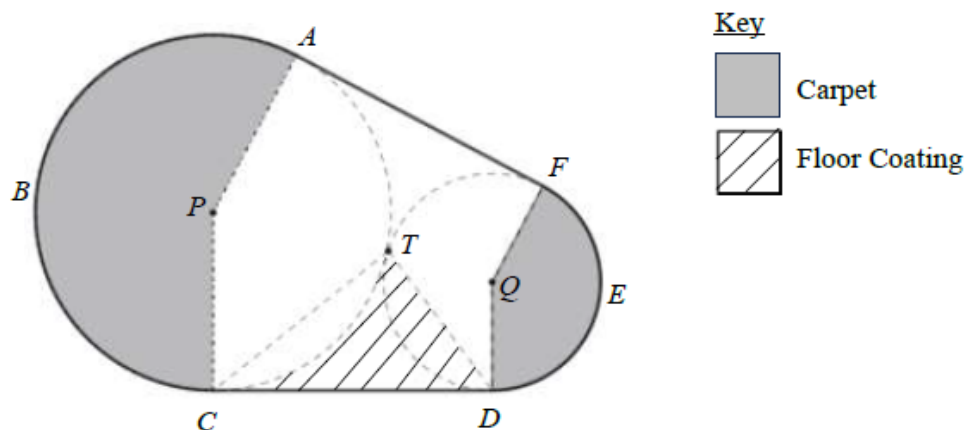
The radius of the smaller circle is 5 m.

(a) Show that the reflex angle $APC = 200.778^\circ$, correct to 3 decimal places.

[2]

Lee wants to turn the room into a showroom.

He intends to cover some areas of the room with carpet and floor coating as shown below.



Lee found information for the supplies he needs on the internet.

Item	Type	Suitable for	Specifications	Cost
Carpet	A	Homes	- Carpet Dimensions: 50 cm × 50 cm - Per carton: 20 pieces - Per carton: 5 m²	\$168 per carton
	B	Offices / Showrooms	- Carpet Dimensions: 50 cm × 50 cm - Per carton: 16 pieces - Per carton: 4 m²	\$158 per carton
Floor Coating	Self-levelling	Homes / Bathrooms / Kitchens		\$50 - \$80 per m ²
	Metallic	Showrooms / Luxury Homes		\$100 - \$150 per m ²
	Anti-slip	Wet Areas / Industrial Floors		\$60 - \$100 per m ²

- (b) Suggest a sensible amount of money that Lee needs to set aside for all his supplies. Justify any decisions you make and show your calculations clearly.

Irene is looking to buy furniture from EKAYA.
She identifies the following items from the catalogue that she needs to purchase, and lists down the approximate amount of time that she will take if she assembles each item.

Item	Retail Price per item (GST included)	Quantity	Approximate time to assemble each item (min)
Bed frame 218 x 155 x 40 cm 36.25 kg	\$249	1	95
Storage solution 12.30 kg	\$119	2	20
Desk with Drawers 140 x 78 x 75 cm 22.10 kg	\$149	1	35
Folding chair 100 x 45 x 47 cm 3.00 kg <i>wall-mountable, requires one hole</i>	\$29	5	10
Bookshelf 80 x 28 x 202 cm 22.50 kg	\$129	1	45
Alarm clock 15 x 13 x 5 cm 0.19 kg	\$19	1	Does not require assembly

(a) Calculate the total cost of the furniture that Irene intends to buy.

[1]

Irene has \$1000 EKAYA credits from lucky draw that allows her to make a one-time purchase on the EKAYA online store.

Irene lives on the 4th floor of an old estate that does not have lift access.
She has back issues and would have difficulties handling weight above 25 kg.
She also wants to wall-mount all the folding chairs, but she does not have a drill.
She does not want to pay more than \$110 out of her pocket.
She does not want to spend more than 80 minutes assembling on her own.

EKAYA

Delivery	Assembly
You can take it home yourself – but you don't have to	You can assemble it yourself – but you don't have to
	
Home delivery is \$35 for orders above 15kg. We'll pick and deliver as many products as you want. Delivery typically takes place within 2-5 days from the date of order. ^ Please note for home delivery, an additional flat rate of \$20 will apply if there is no lift access, or if items cannot fit in the lift at the delivery address located in a building from the 3rd floor onwards. This service is applicable up to the 5th floor only.	Service fees 1. All other furniture that requires assembly: 20% of the original product's retail price. 2. Drilling services are included as part of the assembly services. 3. A \$5 charge per drilled hole applies to products, which require to be wall-mounted and where assembly service has not been booked.

All costs are subject to 9 % GST.

- (b) Suggest how Irene could minimise both the costs spent and the time for assembly for all the furniture purchased. Justify all the decisions you make and show your calculations clearly.

A packing company uses robotic arms to pick up trays from a moving conveyor belt.



A robotic arm consists of two segments connected by joints.

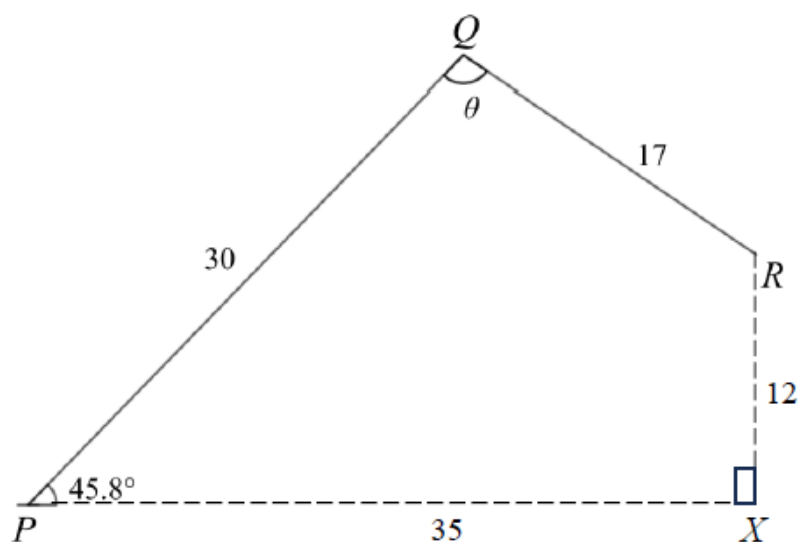
The length of the upper arm PQ is 30 cm.

At P , the shoulder joint is fixed at an angle of 45.8° .

The length of the forearm QR is 17 cm.

At Q , the elbow joint rotates at an angle θ .

The robotic arm picked a tray at point R , which lies 35 cm horizontally and 12 cm vertically above X .



(a) Calculate angle θ , that allows for the arm to pick the tray at R . [2]

(b) Due to calibration errors, there may be a 1° change in θ .
Show that a 1° change in θ will lead to 0.297 cm difference from the intended position of R . [1]

- (c) The company is deciding whether to adopt a new system that uses camera feedback to calculate the joint angle θ in real time.

They are considering two new systems B and C.

System	Description
System A (Original)	Uses pre-programmed joint angle θ . There is a 40% chance that a 1° error in θ will occur, which causes the arm to not be able to pick up the tray.
System B (Real-time)	Uses real-time camera feedback to recalculate θ for every tray. The angle is always accurate.
System C (Hybrid)	Uses pre-programmed joint angle θ . When a 1° error in θ is detected, it uses real-time camera feedback to recalculate the angle θ .

Additional notes:

- Using the original system, the robot handles 300 trays per hour.
- Each successful tray earns the company \$0.50 revenue.
- A 1° error in θ leads to a miss, as the arm cannot pick up the tray.
- Each missed tray incurs a \$0.30 cost.
- Using the camera feedback introduces a 0.4 second delay per tray.
- There is a fixed monthly cost of \$12 000 for using the camera feedback.
- The company operates 160 hours per month.

Determine which system the company should adopt.

Justify the decision you make and show your calculations clearly.

[7]

GM Secondary School Committee plans to install solar panels to reduce electricity cost. The school uses an average of 1000 kWh of electricity per day. The budget allocated to install solar panels is \$200 000.

Two buildings were found suitable for installing the solar panels - Classroom block and Indoor Sports Hall (ISH).

The committee did some research and found the following information.

Flat Panel	Area per Panel (m ²)	Power Output per Panel (kWh/day)	Efficiency (%)	Cost per Panel (including GST)
Panel A	2.0	3.2	19%	\$450
Panel B	1.5	2.4	20%	\$360
Panel C	1.2	1.9	17%	\$280

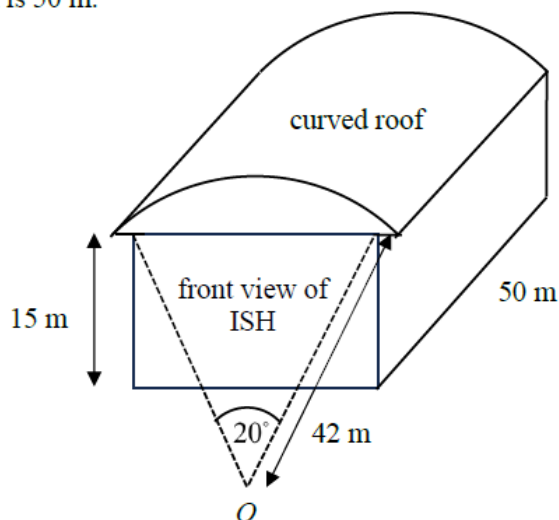
Curved Panel	Cost Per Panel (including GST)
Panel A	Additional 10% of cost of flat panel of the same type
Panel B	
Panel C	

	Installation Fees	
Panel Type	Flat Panel	Curved Panel
Panel A	10% of cost of panel	15% of cost of panel
Panel B	\$40 per panel	\$50 per panel
Panel C	\$35 per panel Free installation for 800 panels and above	\$40 per panel Free installation for 800 panels and above

Classroom block has a flat rectangular roof with dimensions 12 m by 60 m.
ISH has a curved roof formed by an arc of a sector with centre O and radius 42 m.
The height of ISH is 15 m and the length is 50 m.

Classroom block has a flat rectangular roof with dimensions 12 m by 60 m.
ISH has a curved roof formed by an arc of a sector with centre O and radius 42 m.
The height of ISH is 15 m and the length is 50 m.

The front view of ISH is shown below.



	(a) Find the area of the roof of the Classroom block. [1] (b) Find the total cost of buying and installing Panel <i>A</i> on the roof of Classroom block. [2] (c) Find the area of the roof of ISH. [1] The committee needs to decide the location to install solar panels. (d) Suggest a location to install the solar panels, taking into consideration the following: • location(s) • solar panel types with its quantity and cost • school's electricity needs and budget Justify the decision you make by showing your calculations clearly. [6] (e) State one assumption made for part (d). [1]

Jen wants to buy a puppy from a pet shop. The puppy cost \$3588.
The agreed terms of purchase is a 30% deposit with the remaining paid by monthly instalments over a year charged at a simple interest of 8% per annum.

- (a) Calculate the amount for each monthly instalment.

[3]

- (b) As a complimentary gift, Jen was given a sample pack of dog food (dry kibbles).

The dry kibbles are in spherical pellets with diameter of 1.5 cm.
Show that the volume of each spherical pellet is 1.77 cm^3 .

Answer

[1]

- (c) Jen plans to keep her puppy's monthly food budget to \$100.
With some research, she narrowed down to two options; wet canned food or dry kibbles. Dry kibbles can be bought online which has a delivery fee of \$4.50 per purchase, whereas wet canned food can only be bought in stores.

The table below shows the price of each type of pet food and its respective nutritional values.

Type of food	Price (\$)	Delivery Charge (\$)	Nutritional Composition (%)
Dry kibbles (per packet)	15.90	4.50	Fat – 39 Protein – 15 Carbohydrate – 42
Wet canned food (per tin)	2.55	0	Fat – 25 Protein – 38 Carbohydrate – 35

Additional Information:

- Each packet of dry kibbles contains 1500 spherical pellets with diameter of 1.5 cm.
- Wet canned food is sold in tins each containing 191 g of pet food.
- The formula for density is $\rho = \frac{m}{v}$, where ρ is density, m is mass (grams) and v is volume (cm^3).
- The density of the dry kibbles is 0.515 g/cm^3 .

- | | |
|--|---|
| | <p>(i) Jen's puppy requires a total of 250 g of food a day, and she only wishes to buy pet food once a month.
Determine which type of pet food she should choose based on her budget.
Justify your decision and show all calculations clearly.</p> |
|--|---|

Solution

1

(a)

$$\frac{120+100+180+200+220+160+140+170+190+240+200+240}{12} = 180 \text{ mm}$$

(b)

$$0.18 \times 180,000,000 \times 0.75 = 24,300,000 \text{ m}^3$$

$$= 24.3 \text{ million m}^3 \quad \begin{array}{l} \text{[M1 for conversion of units]} \\ \text{[M1]} \end{array}$$

(ci)

Option C as desalination volume > 70 million m³

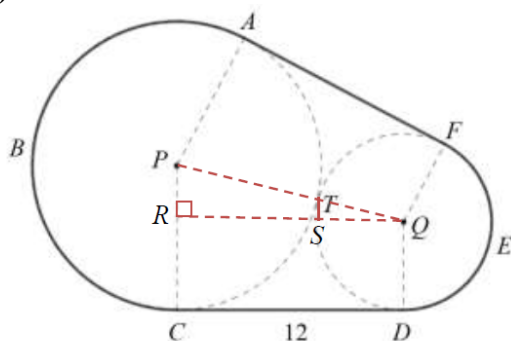
Option D as total volume of NEWater and desalinated water < 85.7 million m³

(cii)

Option	Production Cost (in \$mil) [1 mark]	Energy Cost (in \$mil) [1 mark]	Desalination Penalty (in \$mil) [1 mark]	Carbon Emission (in mil kg)	Carbon Penalty (in \$mil) [1 mark]	Total Cost (in \$mil) [1 mark]
A	$(28 \times 1.30) + (57.7 \times 2.50) = 36.4 + 144.25 = 180.65$	$(28 \times 0.5 + 57.7 \times 2.5) \times 0.26 = (14 + 144.25) \times 0.26 = 41.15$	$(57.7 - 50) \times 0.50 = 7.7 \times 0.50 = 3.85$	$(28 \times 0.2) + (57.7 \times 1.5) = 5.6 + 86.55 = 92.15$	$92.15 \times 1.00 = 92.15$	$180.65 + 41.15 + 3.85 + 92.15 = 317.80$
B	$(30 \times 1.30) + (55.7 \times 2.50) = 39 + 139.25 = 178.25$	$(30 \times 0.5 + 55.7 \times 2.5) \times 0.26 = (15 + 139.25) \times 0.26 = 40.11$	$(55.7 - 50) \times 0.50 = 5.7 \times 0.50 = 2.85$	$(30 \times 0.2) + (55.7 \times 1.5) = 6 + 83.55 = 89.55$	$89.55 \times 0.50 = 44.78$	$178.25 + 40.11 + 2.85 + 44.78 = 265.99$

[1] for choosing Option B with justification

(a)



$$PR = 7.2 - 5 = 2.2 \text{ m}, \quad PQ = 7.2 + 5 = 12.2 \text{ m}$$

$$\cos \angle RPQ = \frac{2.2}{12.2}$$

$$\angle RPQ = \cos^{-1} \left(\frac{2.2}{12.2} \right)$$

$$\approx 79.6111^\circ$$

$$\therefore \text{Reflex } \angle APC = 360^\circ - 2(79.6111^\circ)$$

$$\approx 200.7778^\circ$$

$$= 200.778^\circ \text{ (3 d.p.) (shown)}$$

(b)

Since the two circles are similar, then

$$\text{Obtuse } \angle FQD = 360^\circ - 200.7778^\circ$$

$$\approx 159.2222^\circ$$

Area of carpet required

$$= \frac{200.7778^\circ}{360^\circ} \times \pi(7.2)^2 + \frac{159.2222^\circ}{360^\circ} \times \pi(5)^2$$

$$\approx 125.566 \text{ m}^2$$

$$RQ = \sqrt{12.2^2 - 2.2^2} = 12 \text{ m}$$

Area of coating required

$$= \frac{1}{2}(12)(5 + 7.2) - \frac{79.6111^\circ}{360^\circ} \times \pi(7.2)^2 - \frac{1}{2}(5)^2 \sin 100.3888^\circ$$

$$= 6(12.2) - 36.0152 - 12.2950$$

$$\approx 24.8898 \text{ m}^2$$

$$\text{No. of cartons of carpet required} = 125.566 \div 4$$

$$\approx 31.3915$$

I suggest Lee purchases the minimum of 32 cartons of Type B carpet to lower his supplies cost.

$$\text{Cost of carpet} = 32 \times 158$$

$$= \$5056$$

Since the cost of the floor coating is \$100 - \$150 / m², I will estimate the cost / m² to be \$150 so that Lee can have sufficient money to cover for the coating.

$$\text{Cost of floor coating} = 150 \times 24.8898$$

$$= \$3733.47$$

$$\text{Total cost of supplies} = 5056 + 3733.47$$

$$= \$8789.47$$

$\therefore$ I suggest Lee sets aside \$10 000 for his supplies in case he needs to purchase more supplies due to defects in his carpet / accidental spillages of coating.

3

(a)

$$249 + 119 \times 2 + 149 + 29 \times 5 + 129 + 19 = \$929$$

(b)

Total cost of furniture: \$929

Since Irene has difficulties handling weight above 25 kg, then both delivery and lifting services are essential.

- Delivery: \$35
- Lifting (no lift access): \$20

In order to minimise cost and time, we can do a cost per min analysis.

Item	Assembly cost (20%)	Time (min)	Cost/min	Remarks
Bedframe	$249 \times 0.2 = 49.80$	95	-	> 80 min → assembly service
Storage solution	$119 \times 0.2 \times 2 = 47.60$	$20 \times 2 = 40$	1.19	
Desk with drawers	$149 \times 0.2 = 29.80$	35	0.851	
Folding Chair	$29 \times 0.2 \times 5 = 29$	$10 \times 5 = 50$	0.58	
Bookshelf	$129 \times 0.2 = 25.80$	45	0.573	

Considering that the cost of drilling (separate) is \$25 + 50 min of self-assembly, paying \$29 for assembly (\$4 more to save 50 min) is recommended: use assembly service for folding chairs.

For the remaining 3 items, we do a breakdown as below:

	Storage and Desk	Desk and Bookshelf	Storage and Bookshelf
Pay for remaining	25.80 (bookshelf)	47.60 (storage)	29.80 (desk)
Time needed	$40 + 35$ = 75 min	$35 + 45$ = 80 min	$40 + 45$ = 85 min > 80

Irene should choose to assemble the storage solutions and desk with drawers by herself. Because she would pay the least and spend the least amount of time to assemble.

In summary,

Services acquired: Delivery (\$35) + Lifting (\$20)

Assembly: Bedframe (\$49.80) + Chairs (\$29) + Bookshelf (\$25.80)

Total spending = $929 + (35 + 20 + 49.8 + 29 + 25.8) \times 1.09 - 1000$

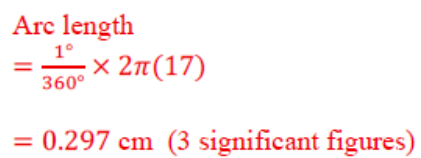
$$= \$102.96 < \$110$$

Time needed = 75 min < 80 min

(a)

$$\angle PQR = \cos^{-1} \left[\frac{30^2 + 17^2 - 37^2}{2(30)(17)} \right]$$

(b)



(c)

	A (Original)	B (Real-time)	C (Hybrid)
Trays/hour	Number of missed trays $= 40\% \times 300$ $= 120$ Number of successful trays $= \frac{60}{100} \times 300$ $= 180$ [M1]	Original time taken for one tray $= \frac{3600}{300} = 12 \text{ s}$ New time taken for each tray (includes delay) $= 12.4 \text{ s}$ Number of successful trays $= \frac{3600}{12.4}$ $= 290$ (nearest whole number) [M1]	Similar to A Successful trays = 180 Total time taken for successful trays $= \frac{3600}{300} \times 180$ $= 2160 \text{ s}$ Includes B Number of trays picked up after recalculation. $= \frac{3600 - 2160}{12.4}$ $= 116$ (nearest whole number) Total number of trays $= 180 + 116$ $= 296$ [M1]
Revenue/hour	$180 \times \$0.50$ $= \$90$	$290 \times \$0.50$ $= \$145$	$296 \times \$0.50$ $= \$148$
Missed cost/hour	$120 \times \$0.30$ $= \$36$	\$0	\$0
Camera cost/hour	\$0	$\frac{\$12000}{160} = \75	$\frac{\$12000}{160} = \75
Net profit/hour	$\$90 - \$36 = \$54$	$\$145 - \75 $= \$70$	$\$148 - \75 $= \$73$
Monthly profit (160 hour / month)	$\$54 \times 160$ $= \$8640$ [M1]	$\$70 \times 160$ $= \$11200$ [M1]	$\$73 \times 160$ $= \$11680$ [M1]

Since System C generates the most profit, the company should adopt System C.

5	a	Area of Classroom Block Roof $= 12 \times 60$ $= 720\text{m}^2$
	b	Number of Panel A to purchase $= 720 \div 2$ $= 360$ Total Cost of Panel A & installation $= 360 \times 450 \times 110\%$ $= \$178\,200$
	c	Arc Length of ISH Rooftop $= \frac{20}{360} \times 2 \times \pi \times 42$ $= \frac{168\pi}{36} \text{m}$ $= \frac{14\pi}{3} \text{m}$ Area of the ISH Rooftop $= \left(\frac{14\pi}{3} \times 50 \right) \text{m}$ $= \frac{700\pi}{3} \text{m}^2$ $= 733.038\text{m}^2$ $= 733\text{m}^2$
	d	Following (b) working, and to repeat working for all 3 panels Maximum Number of Panel A to purchase $= 360$ Total Power Output $= 360 \times 3.2$ $= 1152\text{kWh}$ Panel A can be installed because it exceeds 1000kWh. Maximum total Cost of Panel A & installation $= \$178,200$

Maximum Number of Panel B to purchase

$$= 720 \div 1.5$$

$$= 480$$

Total Power Output

$$= 480 \times 2.4$$

$$= 1152 \text{ kWh}$$

Panel B can be installed because it exceeds 1000 kWh.

Maximum Total Cost of Panel B & installation

$$= 480 \times (360 + 40)$$

$$= \$192,000$$

Maximum Number of Panel C to purchase

$$= 720 \div 1.2$$

$$= 600$$

Total Power Output

$$= 600 \times 1.9$$

$$= 1140 \text{ kWh}$$

Maximum Total Cost of Panel C & installation

$$= 600 \times (280 + 35)$$

$$= \$189,000$$

ISH

Since the total area of ISH rooftop is bigger than the total area of Classroom Block rooftop, then all 3 panels will also be able to meet and exceed the power output.

At this point, solutions may vary.

1) Students can opt to have the maximum installation, and to sell back into the grid for \$.

Assumption: school can generate income too if they produce more than enough power. Generation income is a bonus to the school.

2) Students can also opt to build “just-enough” for the power supply, since both locations have more than enough land space.

Students can also do a hybrid of panels. Example. 200 panels A, 100 panels B etc.

Possible Solution henceforth.

(Using the cheapest approach)

The ISH will be out of question since the area is greater, and there is additional cost for curved panel. (Hence, it is no longer necessary to calculate the cost for installing on ISH roof.)

But since it is to find the cheapest, then students have to find the minimum number of panels to buy. They cannot use the cost above since it is the maximum cost. (and unless or otherwise, they explain their rationale for above).

(Using the maximum area approach)

Students must then use ISH rooftop. Then they will have to find the 3 panels cost and justify accordingly.

Note that the ISH area is rounded off. The number of panels is strictly less than the area of ISH rooftop.

(Using any other possible/believable reasons, such as it is “nicer” to build on top of ISH, to minimise disruption in classes, reflection will be glaring etc)

Marks will be allocated according.

Most importantly, their working should be clear, with good annotation, and to write one's conclusion accordingly.

Sample Solution:

Following (b) working, and to repeat working for all 3 panels,

Maximum Number of Panel A to purchase

$$= 360$$

Total Power Output

$$= 360 \times 3.2$$

$$= 1152 \text{ kWh}$$

Panel A can be installed because it exceeds 1000kWh.

$$\begin{aligned} \text{Minimum Number of panels} &= 1000 \div 3.2 \\ &= 313 \end{aligned}$$

Minimum Total Cost of Panel A & installation

$$= \$ (313 \times 450) \times 1.1$$

$$= \$154,935$$

Maximum Number of Panel B to purchase

$$= 720 \div 1.5$$

$$= 480$$

Total Power Output

$$= 480 \times 2.4$$

$$= 1152 \text{ kWh}$$

Panel B can be installed because it exceeds 1000kWh.

$$\begin{aligned} \text{Minimum Number of panels} &= 1000 \div 2.4 \\ &= 417 \end{aligned}$$

Minimum Total Cost of Panel B & installation

$$= \$ (417 \times 360) \times 1.1$$

$$= \$165,132$$

Maximum Number of Panel C to purchase

$$= 720 \div 1.2$$

$$= 600$$

Total Power Output

$$= 600 \times 1.9$$

$$= 1140 \text{ kWh}$$

Panel C can be installed because it exceeds 1000kWh.

Minimum Number of panels

$$= 1000 \div 1.9$$

$$= 526.315789$$

$$= 527$$

Minimum Total Cost of Panel C & installation

$$= \$ (527 \times 280) \times 1.1$$

$$= \$162,316$$

ISH will not be considered because the curved panels are more expensive than flat panels. There won't be a need to do any calculation since it will be definitely more panels for all panels.

Solution

Location: Classroom Block Rooftop

Panel : A

Quantity : 313

This solution will provide just enough power for the school at the least cost.

e | Assumption: school has tight budget; be prudent etc

(a)	Per month instalment = $\frac{3588(0.7)(1.08)}{12}$
	=226.04
	(b) $\frac{4}{3}\pi(0.75^3) = 1.7671$
	=1.77 (shown)
	(c) Total food puppy needs per month, $250 \times 30 = 7500$
	No. of tins of wet food needed, $7500 \div 191 = 39.267$
	=40 (rounded up)
	Total price of wet food, $40 \times 2.55 = 102$
	Volume of kibbles in a packet, $1500 \times 1.7671 = 2650.65$
	Mass of kibbles in a packet, $2650.65 \times 0.515 = 1351.8315$
	No. of packets of kibbles needed, $7500 \div 1351.8315 = 5.548$
	=6 packets (rounded up)
	Price of kibbles + delivery, $6 \times 15.90 + 4.50 = 99.90$

Jen should get the dry kibbles as it is within her budget of \$100.